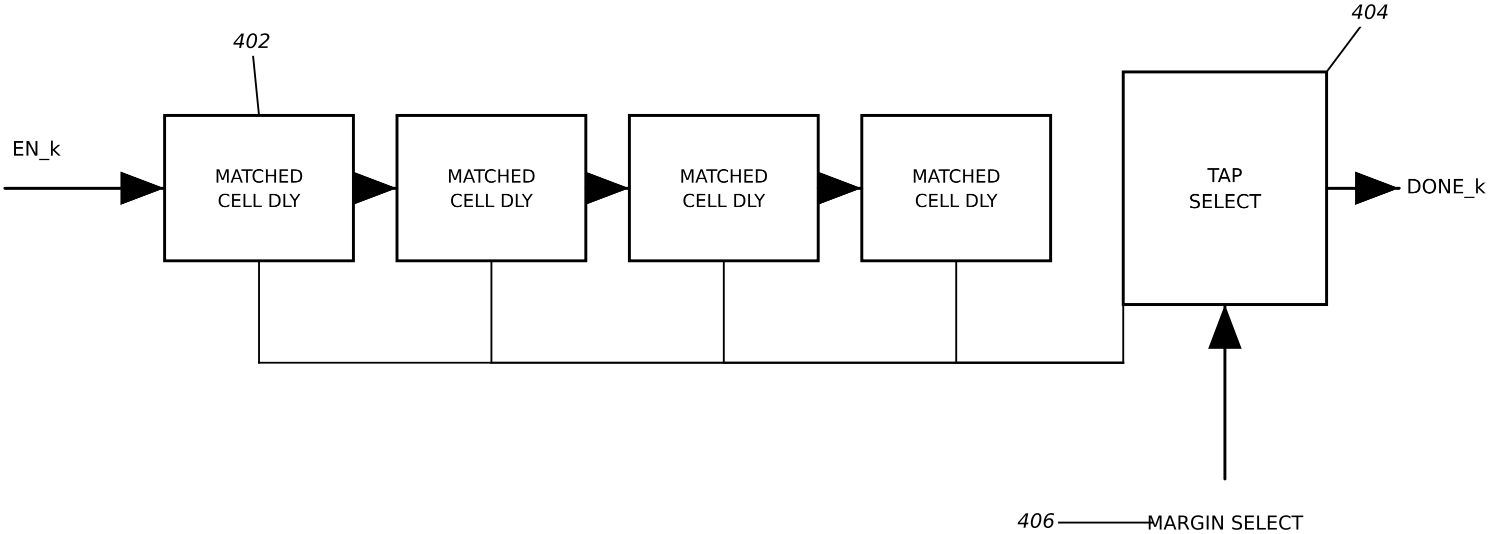
PROGRAMMABLE MATCHED-DELAY COMPLETION GENERATOR



SIZING CONSTRAINT

selected delay $\geq$ $t_{pd,max}$ (class k) + guard band

A tap shorter than the worst-case combinational propagation delay through the class asserts DONE_k early, and the next class then evaluates on inputs that have not settled.

That premature evaluation is the perturbation the annealing schedule exploits: undersized taps early in the schedule, full-width taps at the end.

The same constraint applied to the node delays of FIG. 6 is a correctness requirement rather than a control variable — a node delay shorter than the propagation delay through its own logic makes the class ordering inoperative, because commits then interleave in an order the delay values no longer determine.

FIG. 4